

Honors Pre-Calculus

Complete Study Notes

Contents

1	Functions and Their Properties	3
1.1	Combining Functions	3
1.2	Increasing, Decreasing, and Symmetry	3
1.3	Transformations of Functions	3
1.4	Inverse Functions	4
2	Polynomial and Rational Functions	4
2.1	Dividing Polynomials	4
2.2	The Remainder & Factor Theorems	4
2.3	Zeros and Complex Roots	4
2.4	End Behavior of Polynomials	4
2.5	Asymptotes of Rational Functions	5
3	Exponential and Logarithmic Functions	5
3.1	Exponential Functions	5
3.2	Logarithms	6
3.3	Graphs of Logarithmic Functions	6
3.4	Properties (Laws) of Logarithms	6
3.5	Applications of Logarithms	7
4	Trigonometry: Angles and the Six Functions	7
4.1	Angle Basics	7
4.2	The Six Trigonometric Ratios	7
4.3	Special Triangles	7
4.4	Reference Angles	7
4.5	Graphs of the Trigonometric Functions	8
5	Analytic Trigonometry: Identities	9
5.1	Fundamental Identities	9

5.2	Sum and Difference Identities	10
5.3	Double-Angle Identities	10
5.4	Power-Reducing Identities	10
5.5	Half-Angle Identities	10
5.6	Laws of Sines and Cosines	10
6	Inverse Trigonometric Functions	11
6.1	Domains and Ranges	11
6.2	Composition	11
7	Vectors in the Plane	11
7.1	Dot Product	12
8	Polar Coordinates	12
9	Conic Sections	12
9.1	The Ellipse	12
9.2	The Hyperbola	13
10	Sequences and Series	14
10.1	Limits of Sequences	14
10.2	Arithmetic Sequences	14
10.3	Geometric Sequences	14
11	Counting: Permutations and Combinations	14
12	Matrices	15
13	Systems of Inequalities and Partial Fractions	15
13.1	Solving a System of Inequalities	15
13.2	Partial Fraction Decomposition	15

1. Functions and Their Properties

1.1 Combining Functions

For functions f and g , the four basic combinations are:

$$(f + g)(x), \quad (f - g)(x), \quad (f \cdot g)(x), \quad \left(\frac{f}{g}\right)(x) \text{ where } g(x) \neq 0.$$

The quotient f/g requires the restriction $\{x : g(x) \neq 0\}$.

1.2 Increasing, Decreasing, and Symmetry

- As x increases, if $f(x)$ **decreases**, the function is *decreasing*.
- As x increases, if $f(x)$ **increases**, the function is *increasing*.
- If $f(x)$ stays the same, the function is *constant*.
- **Even symmetry:** symmetric about the y -axis, $f(-x) = f(x)$.
- **Odd symmetry:** symmetric about the origin, $f(-x) = -f(x)$.

1.3 Transformations of Functions

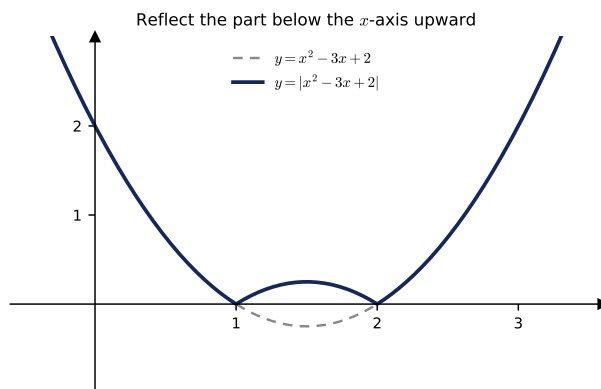
- **Shifts (translations):** adding inside/outside the function shifts the graph (rigid translation).
- **Stretches / shrinks:** a coefficient changes the shape, e.g. $y = ax^2 + bx + c$.
- **Reflections:** flip the graph across an axis.

	$f(x)$ (vertical)	$f\left(\frac{x}{c}\right)$ (horizontal)
$c > 1$	vertical stretch	horizontal stretch
$0 < c < 1$	vertical shrink	horizontal shrink

Example (absolute value transformation). Starting from $y = x^2 - 3x + 2$:

$y = |x^2 - 3x + 2|$ reflects the portion below the x -axis upward;

$-y = |x|^2 - 3|x| + 2$ applies symmetry in x and reflects vertically.



Taking the absolute value reflects the negative portion of the parabola above the x -axis.

1.4 Inverse Functions

$f^{-1}(y) = x \iff f(x) = y$. The graph of f^{-1} is the reflection of f across the line $y = x$.

	$f(x)$	$f^{-1}(x)$
Domain	A	B
Range	B	A

The domain and range swap between a function and its inverse.

2. Polynomial and Rational Functions

2.1 Dividing Polynomials

Two methods are used to divide polynomials:

1. **Synthetic division** (short division) — quick when dividing by a linear factor.
2. **Long division** — general method, also used to find slant (oblique) asymptotes.

2.2 The Remainder & Factor Theorems

To test whether $(x - a)$ is a factor of $P(x)$, evaluate $P(a)$:

$$x - a = 0 \Rightarrow x = a, \quad P(a) = 0 \iff (x - a) \text{ is a factor (remainder} = 0).$$

2.3 Zeros and Complex Roots

- Complex roots occur in conjugate pairs: $x = a + bi$ and $x = a - bi$.
- If the degree is **odd**, the number of real zeros must be odd; if the degree is **even**, the number of real zeros is even.
- Use long division to find a slant asymptote.
- Domain $\Rightarrow$ restrictions on x ; Range $\Rightarrow$ restrictions on y .

2.4 End Behavior of Polynomials

For a polynomial P with leading term as indicated:

Degree	Leading coeff.	Left end	Right end
odd	+	falls ($\downarrow$)	rises ($\uparrow$)
odd	−	rises ($\uparrow$)	falls ($\downarrow$)
even	+	rises ($\uparrow$)	rises ($\uparrow$)
even	−	falls ($\downarrow$)	falls ($\downarrow$)

2.5 Asymptotes of Rational Functions

Let

$$r(x) = \frac{a_n x^n}{a_m x^m} \quad (\text{leading terms of numerator and denominator}).$$

Horizontal asymptotes (H.A.):

- $n > m$: no horizontal asymptote (an oblique asymptote may exist).
- $n = m$: horizontal asymptote $y = \frac{a_n}{a_m}$.
- $n < m$: horizontal asymptote $y = 0$.

Vertical asymptotes (V.A.): set the denominator = 0 and solve for x .

Oblique (slant) asymptote: occurs when $\deg(\text{numerator}) = \deg(\text{denominator}) + 1$; found by *long division*.

3. Exponential and Logarithmic Functions

3.1 Exponential Functions

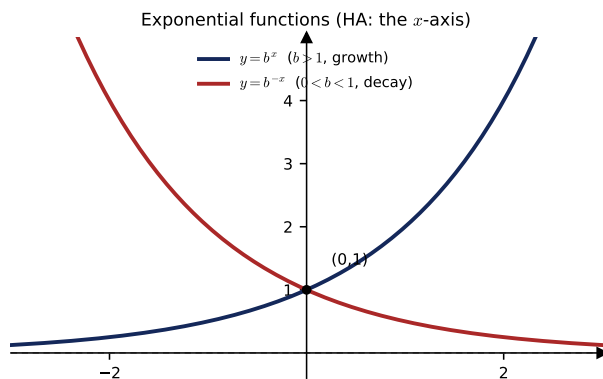
The exponential function is $y = b^x$ (or $y = b^{-x}$).

- $0 < b < 1$: exponential **decay**.
- $b > 1$: exponential **growth**.

For $y = b^x$ with $b > 1$:

- Domain $(-\infty, \infty)$
- Range $(0, \infty)$
- y -intercept $(0, 1)$
- Increasing
- The x -axis is a horizontal asymptote
- Continuous everywhere

For $y = b^{-x}$ the function is decreasing (a reflection across the y -axis).



Growth ($b > 1$) versus decay ($0 < b < 1$); the x -axis is a horizontal asymptote.

The natural base: $e \approx 2.718$. Compound growth: $P(1 + r)^t$.

3.2 Logarithms

The logarithm is the **inverse function** of the exponential function. Given $a > 0$, $a \neq 1$:

$$\log_a 1 = 0, \quad \log_a a = 1, \quad \log_a a^x = x, \quad a^{\log_a x} = x.$$

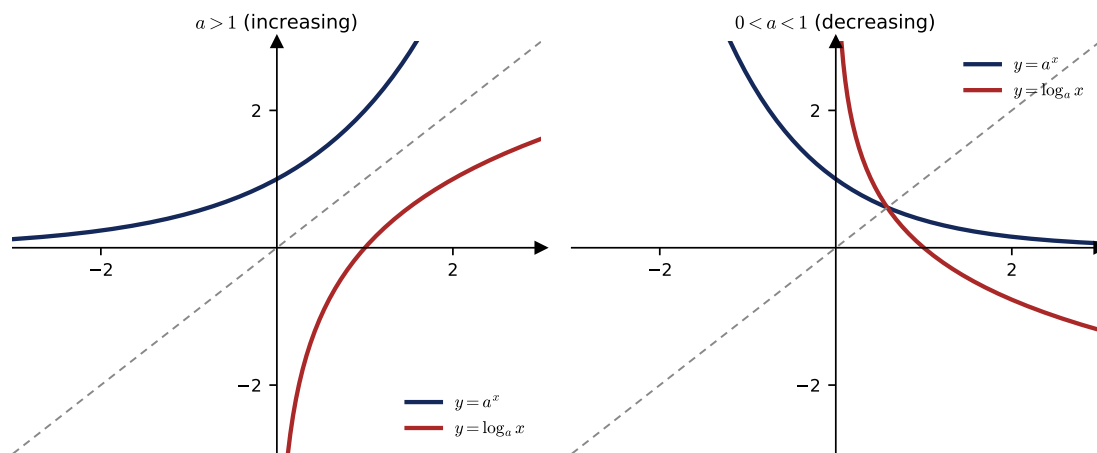
Special bases:

- Base e — natural log $\ln$: $\ln e = 1$, $\ln 1 = 0$, $\ln e^x = x$, $e^{\ln x} = x$.
- Base 10 — common log $\log$: $\log 1 = 0$, $\log 10 = 1$, $\log 10^x = x$, $10^{\log x} = x$.

3.3 Graphs of Logarithmic Functions

The graphs of $y = a^x$ and $y = \log_a x$ are reflections of each other across $y = x$.

- For $0 < a < 1$: $\log_a x$ is decreasing, passing through $(1, 0)$.
- For $a > 1$: $\log_a x$ is increasing, passing through $(1, 0)$.



$y = a^x$ and $y = \log_a x$ are mirror images across the line $y = x$.

3.4 Properties (Laws) of Logarithms

For $a, b > 0$ and $\neq 1$:

$$\begin{aligned} \log_b(M \cdot N) &= \log_b M + \log_b N, & \log_b\left(\frac{M}{N}\right) &= \log_b M - \log_b N, \\ \log_b a^c &= c \log_b a, & \log_a b &= \frac{\log_c b}{\log_c a} \text{ (change of base, } c > 0, c \neq 1). \end{aligned}$$

Extended results:

$$\log_{a^m} b^n = \frac{n}{m} \log_a b, \quad \log_a b = \frac{1}{\log_b a}.$$

3.5 Applications of Logarithms

$$\begin{array}{ll} \text{pH} = -\log[\text{H}^+] & \text{(acidity)} \\ T = T_m + (T_0 - T_m)e^{-kt} & \text{(Newton's law of cooling)} \\ u = \log\left(\frac{a}{a_0}\right) & \text{(Richter scale)} \end{array}$$

4. Trigonometry: Angles and the Six Functions

4.1 Angle Basics

- $1 \text{ rad} = \frac{180^\circ}{\pi} \approx 57.3^\circ$.
- An angle has an *initial side* and a *terminal side*.
- **Positive direction:** counterclockwise.
- **Quadrantal angle:** terminal side lies on an axis.
- **Periodic:** angles repeat every 2π (or 360°).
- **Coterminal angles:** same terminal side, different measure: $\beta = \alpha + 360^\circ k$ (with $k = 0, \beta = \alpha$).

Sub-divisions of a degree: $1^\circ = 60'$ (minutes), $1' = 60''$ (seconds).

4.2 The Six Trigonometric Ratios

$$\sin = \frac{\text{opp}}{\text{hyp}}, \quad \cos = \frac{\text{adj}}{\text{hyp}}, \quad \tan = \frac{\text{opp}}{\text{adj}}, \quad \csc = \frac{\text{hyp}}{\text{opp}}, \quad \sec = \frac{\text{hyp}}{\text{adj}}, \quad \cot = \frac{\text{adj}}{\text{opp}}.$$

4.3 Special Triangles

$30^\circ-60^\circ-90^\circ$:

$$\begin{array}{ll} \sin 30^\circ = \frac{1}{2} = \cos 60^\circ, & \csc 30^\circ = 2 = \sec 60^\circ, \\ \sin 60^\circ = \frac{\sqrt{3}}{2} = \cos 30^\circ, & \csc 60^\circ = \frac{2}{\sqrt{3}} = \sec 30^\circ, \\ \tan 30^\circ = \frac{1}{\sqrt{3}} = \cot 60^\circ, & \cot 30^\circ = \sqrt{3} = \tan 60^\circ. \end{array}$$

$45^\circ-45^\circ-90^\circ$:

$$\sin 45^\circ = \frac{\sqrt{2}}{2} = \cos 45^\circ, \quad \csc 45^\circ = \sqrt{2} = \sec 45^\circ, \quad \tan 45^\circ = 1 = \cot 45^\circ.$$

4.4 Reference Angles

The **reference angle** is the acute angle bounded by the terminal side and the x -axis.

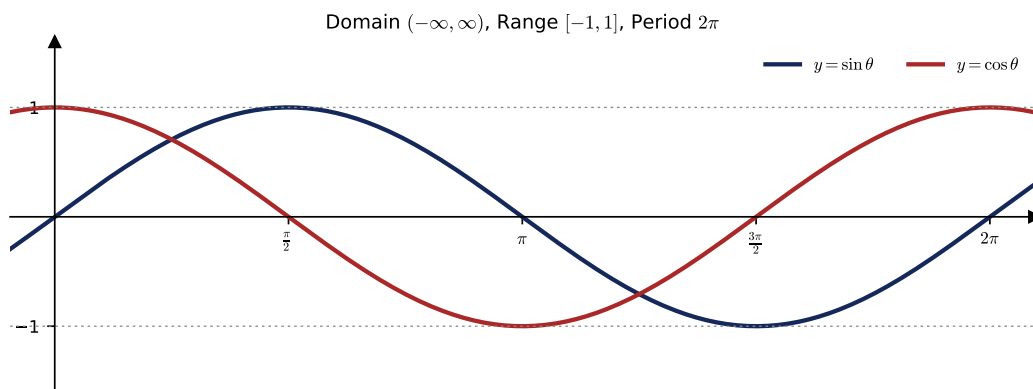
Quadrant of terminal side	Reference angle θ_R
QI	$\theta_R = \theta$
QII	$\theta_R = 180^\circ - \theta$
QIII	$\theta_R = \theta - 180^\circ$
QIV	$\theta_R = 360^\circ - \theta$

Periodicity: $\sin(\theta + 2k\pi) = \sin \theta$ and $\cos(\theta + 2k\pi) = \cos \theta$.

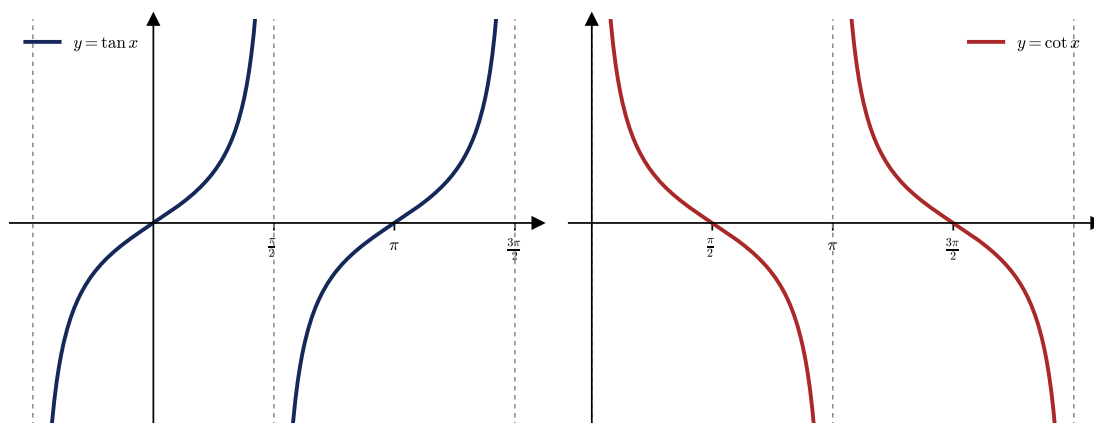
4.5 Graphs of the Trigonometric Functions

Function	Domain	Range	Period
$y = \sin \theta$	$(-\infty, \infty)$	$[-1, 1]$	2π
$y = \cos \theta$	$(-\infty, \infty)$	$[-1, 1]$	2π
$y = \tan x$	$x \neq \frac{\pi}{2} + k\pi$	$(-\infty, \infty)$	π
$y = \cot x$	$x \neq k\pi$	$(-\infty, \infty)$	π
$y = \sec x$	$x \neq \frac{\pi}{2} + k\pi$	$(-\infty, -1] \cup [1, \infty)$	2π
$y = \csc x$	$x \neq k\pi$	$(-\infty, -1] \cup [1, \infty)$	2π

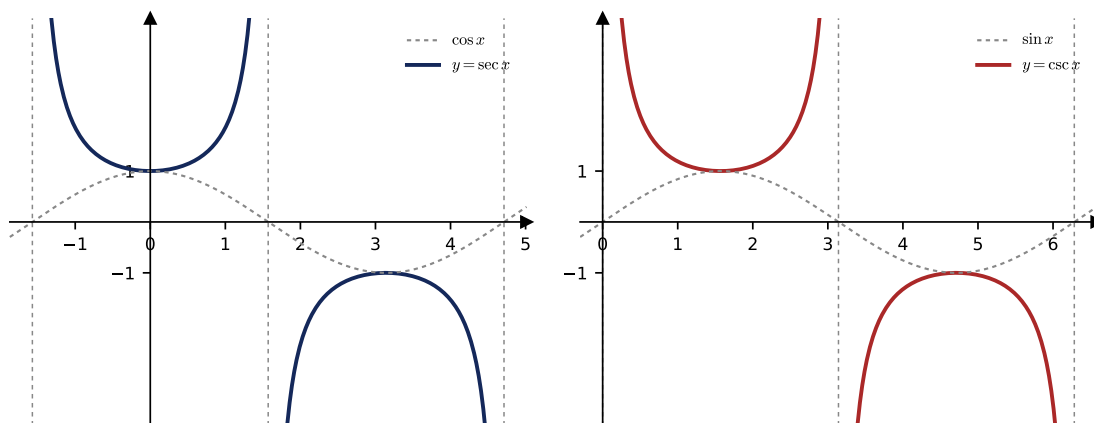
Note $\sec x = \frac{1}{\cos x}$ and $\csc x = \frac{1}{\sin x}$.



The sine and cosine curves (cosine is sine shifted left by $\frac{\pi}{2}$).



Tangent and cotangent, each with period π and vertical asymptotes.



Secant and cosecant (dashed: the reciprocal cosine/sine for reference).

For a general sinusoid $y = a \sin(bx)$, the period is $T = \frac{2\pi}{\omega}$ where $\omega = b$.

5. Analytic Trigonometry: Identities

5.1 Fundamental Identities

Reciprocal:

$$\sin \theta = \frac{1}{\csc \theta}, \quad \cos \theta = \frac{1}{\sec \theta}, \quad \tan \theta = \frac{1}{\cot \theta}, \quad \cot \theta = \frac{1}{\tan \theta}, \quad \sec \theta = \frac{1}{\cos \theta}, \quad \csc \theta = \frac{1}{\sin \theta}.$$

Quotient:

$$\tan \theta = \frac{\sin \theta}{\cos \theta}, \quad \cot \theta = \frac{\cos \theta}{\sin \theta}.$$

Pythagorean:

$$\sin^2 \theta + \cos^2 \theta = 1, \quad \tan^2 \theta + 1 = \sec^2 \theta, \quad 1 + \cot^2 \theta = \csc^2 \theta.$$

Cofunction:

$$\begin{aligned} \sin\left(\frac{\pi}{2} - \theta\right) &= \cos \theta, & \cos\left(\frac{\pi}{2} - \theta\right) &= \sin \theta, \\ \tan\left(\frac{\pi}{2} - \theta\right) &= \cot \theta, & \cot\left(\frac{\pi}{2} - \theta\right) &= \tan \theta, \\ \sec\left(\frac{\pi}{2} - \theta\right) &= \csc \theta, & \csc\left(\frac{\pi}{2} - \theta\right) &= \sec \theta. \end{aligned}$$

Even–Odd:

$$\begin{aligned} \sin(-\theta) &= -\sin \theta \text{ (odd)}, & \cos(-\theta) &= \cos \theta \text{ (even)}, & \tan(-\theta) &= -\tan \theta \text{ (odd)}, \\ \csc(-\theta) &= -\csc \theta \text{ (odd)}, & \sec(-\theta) &= \sec \theta \text{ (even)}, & \cot(-\theta) &= -\cot \theta \text{ (odd)}. \end{aligned}$$

5.2 Sum and Difference Identities

$$\begin{aligned}\sin(x \pm y) &= \sin x \cos y \pm \cos x \sin y, \\ \cos(x \pm y) &= \cos x \cos y \mp \sin x \sin y, \\ \tan(x \pm y) &= \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}.\end{aligned}$$

5.3 Double-Angle Identities

$$\begin{aligned}\sin(2u) &= 2 \sin u \cos u, \\ \cos(2u) &= \cos^2 u - \sin^2 u = 2 \cos^2 u - 1 = 1 - 2 \sin^2 u.\end{aligned}$$

Worked check. Solving $\sin 2x = \cos x$:

$$2 \sin x \cos x = \cos x \Rightarrow \cos x(2 \sin x - 1) = 0.$$

5.4 Power-Reducing Identities

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}, \quad \tan^2 u = \frac{1 - \cos 2u}{1 + \cos 2u}.$$

5.5 Half-Angle Identities

$$\begin{aligned}\sin^2 \frac{u}{2} &= \frac{1 - \cos u}{2}, & \cos^2 \frac{u}{2} &= \frac{1 + \cos u}{2}, & \tan^2 \frac{u}{2} &= \frac{1 - \cos u}{1 + \cos u}.\end{aligned}$$

$$\sin \frac{u}{2} = \pm \sqrt{\frac{1 - \cos u}{2}}, \quad \cos \frac{u}{2} = \pm \sqrt{\frac{1 + \cos u}{2}}, \quad \tan \frac{u}{2} = \pm \sqrt{\frac{1 - \cos u}{1 + \cos u}}.$$

The sign (+ or -) is determined by the quadrant of $\frac{u}{2}$.

5.6 Laws of Sines and Cosines

Law of Sines:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}.$$

Law of Cosines:

$$\begin{aligned}a^2 &= b^2 + c^2 - 2bc \cos A, \\ b^2 &= a^2 + c^2 - 2ac \cos B, \\ c^2 &= a^2 + b^2 - 2ab \cos C,\end{aligned}$$

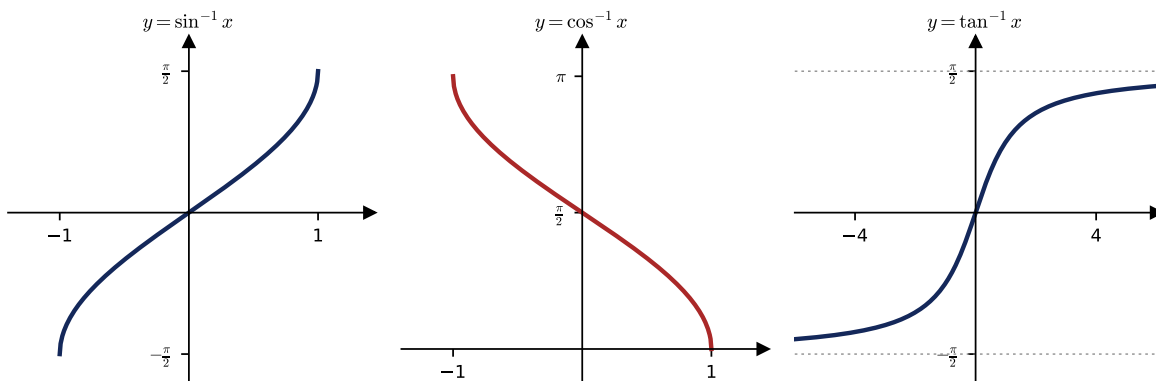
equivalently $\cos A = \frac{b^2 + c^2 - a^2}{2bc}.$

6. Inverse Trigonometric Functions

The six inverse functions are $\sin^{-1} x$, $\cos^{-1} x$, $\tan^{-1} x$, $\cot^{-1} x$, $\sec^{-1} x$, $\csc^{-1} x$. For the inverse functions the independent variable is a ratio and the dependent variable is an angle (the reverse of the forward functions).

6.1 Domains and Ranges

Inverse function	Domain	Range
$\sin^{-1} x$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
$\cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1} x$	$(-\infty, \infty)$	$(-\frac{\pi}{2}, \frac{\pi}{2})$



The three main inverse functions; note the restricted ranges that make them functions.

6.2 Composition

For x in the appropriate domain:

$$\sin(\sin^{-1} x) = x, \quad \sin^{-1}(\sin x) = x; \quad \cos(\cos^{-1} x) = x, \quad \cos^{-1}(\cos x) = x;$$

$$\tan(\tan^{-1} x) = x, \quad \tan^{-1}(\tan x) = x.$$

The value x must lie in the domain for these to hold.

7. Vectors in the Plane

A vector has **magnitude** and **direction**, written with angle brackets: $\vec{v} = \langle a, b \rangle$, where a is the horizontal and b the vertical component.

$$\begin{aligned} \frac{y}{x} &= \tan \theta, & \theta &= \tan^{-1}\left(\frac{b}{a}\right), \\ a &= |\vec{v}| \cos \theta, & b &= |\vec{v}| \sin \theta. \end{aligned}$$

Zero vector: $\langle 0, 0 \rangle$, with $|\vec{u}| = 0$ — no direction, no motion.

Operations (with scalar k):

$$\vec{u} + \vec{v} = \langle u_1 + v_1, u_2 + v_2 \rangle, \quad k\vec{u} = \langle ku_1, ku_2 \rangle.$$

Standard unit vectors: $\vec{i} = \langle 1, 0 \rangle$, $\vec{j} = \langle 0, 1 \rangle$.

Unit vector in the direction of $\vec{u}$: $\hat{a} = \frac{\vec{u}}{|\vec{u}|}$.

7.1 Dot Product

For $\vec{u} = \langle u_1, u_2 \rangle$ and $\vec{v} = \langle v_1, v_2 \rangle$:

$$\vec{u} \cdot \vec{v} = u_1v_1 + u_2v_2.$$

Relationships between two vectors:

- $\vec{u} \cdot \vec{v} = 0 \Rightarrow \vec{u} \perp \vec{v}$ (orthogonal).
- $\vec{u} = k\vec{v}$ ($k \neq 0$) $\Rightarrow \vec{u} \parallel \vec{v}$ (parallel).
- Otherwise they intersect at angle θ where $\cos \theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}| |\vec{v}|}$.

Also $|\vec{u}|^2 = \vec{u} \cdot \vec{u}$.

8. Polar Coordinates

A point is described by (r, θ) instead of (x, y) :

$$x = r \cos \theta, \quad y = r \sin \theta, \quad x^2 + y^2 = r^2.$$

A circle of radius r centered at (h, k) : $(x - h)^2 + (y - k)^2 = r^2$.

Example (convert to rectangular).

$$r = -3 \sin \theta \Rightarrow r^2 = -3r \sin \theta \Rightarrow x^2 + y^2 = -3y.$$

9. Conic Sections

The conic sections (parabola, ellipse, hyperbola) arise from slicing a double cone.

9.1 The Ellipse

Definition: the set of points P with $PF_1 + PF_2 = 2a$, where $F_1F_2 = 2c$ and $a > c$.

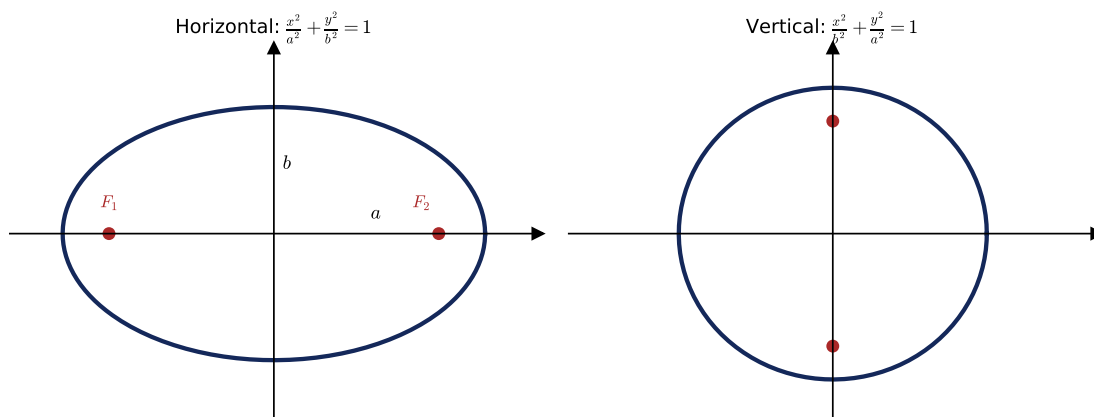
$$\text{Eccentricity } e = \frac{c}{a}.$$

Standard equations (center (h, k)):

$$\text{Horizontal major axis } (a > b) : \frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1,$$

$$\text{Vertical major axis } (a > b) : \frac{(x - h)^2}{b^2} + \frac{(y - k)^2}{a^2} = 1.$$

- Semi-major axis a , semi-minor axis b .
- Pythagorean relation: $a^2 = b^2 + c^2$, so foci distance $= \sqrt{a^2 - b^2}$.
- Vertices: $(h \pm a, k)$ or $(h, k \pm a)$; Foci: $(h \pm c, k)$ or $(h, k \pm c)$.
- Perihelion (closest) $= a - c$; Aphelion (farthest) $= a + c$.



Ellipses with horizontal and vertical major axes; red dots are the foci.

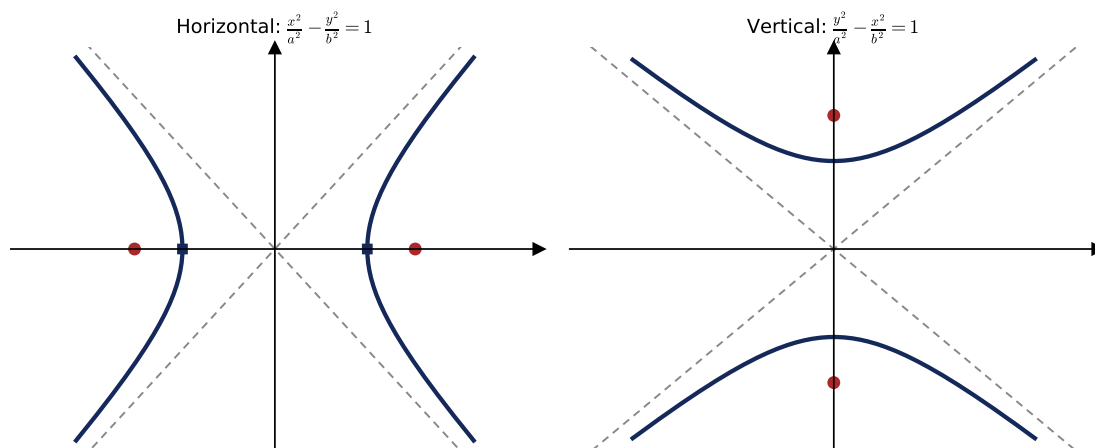
9.2 The Hyperbola

Definition: the set of points with $|PF_1 - PF_2| = 2a$, with $c^2 = a^2 + b^2$ and $e > 1$.

$$\text{Horizontal: } \frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1, \quad \text{Vertical: } \frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1.$$

	Focal axis = x	Focal axis = y
Foci	$(\pm c, 0)$	$(0, \pm c)$
Vertices	$(\pm a, 0)$	$(0, \pm a)$
Semi-transverse axis	a	a
Semi-conjugate axis	b	b
Relation	$c^2 = a^2 + b^2$	$c^2 = a^2 + b^2$
Asymptotes	$y = \pm \frac{b}{a}(x-h) + k$	$y = \pm \frac{a}{b}(x-h) + k$

For the hyperbola, foci distance $= \sqrt{a^2 + b^2}$.



Hyperbolas with their asymptotes (dashed); squares mark vertices, dots mark foci.

10. Sequences and Series

10.1 Limits of Sequences

$\lim_{n \rightarrow \infty} a_n = L$ (a specific number) $\Rightarrow$ the sequence **converges**.

For example $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$. If the limit does not exist, the sequence **diverges**.

10.2 Arithmetic Sequences

$$a_n = a_1 + (n - 1)d, \quad S_n = \frac{(a_1 + a_n)n}{2} = na_1 + \frac{n(n - 1)}{2}d.$$

10.3 Geometric Sequences

$$a_n = a_1 r^{n-1}, \quad S_n = \frac{a_1(1 - r^n)}{1 - r}.$$

Derivation of S_n :

$$\begin{aligned} S_n &= a_1 + a_1 r + a_1 r^2 + \cdots + a_1 r^{n-1}, \\ rS_n &= a_1 r + a_1 r^2 + \cdots + a_1 r^n, \\ S_n - rS_n &= a_1 - a_1 r^n \Rightarrow S_n = \frac{a_1(1 - r^n)}{1 - r}. \end{aligned}$$

11. Counting: Permutations and Combinations

Permutations — arranging members *in order*:

$${}_nP_r = \frac{n!}{(n - r)!}.$$

Combinations — selecting items from a collection, *order does not matter*:

$${}_nC_r = \binom{n}{r} = \frac{{}_nP_r}{r!} = \frac{n!}{r!(n-r)!}.$$

Permutations with repeated items:

$$P = \frac{n!}{n_1! n_2! n_3! \cdots n_k!}.$$

12. Matrices

Dimensions are written (rows $\times$ columns): a 3×3 matrix has 9 entries, a 2×3 matrix has 6 entries. For dimension $a \times b$, a is the number of rows and b the number of columns. The entry $g_{i,j}$ sits in row i , column j ; e.g. for

$$G = \begin{bmatrix} 4 & 14 & 7 \\ 13 & 5 & 13 \\ -20 & 4 & 22 \end{bmatrix}, \quad g_{2,1} = 13.$$

Scalar multiplication multiplies every entry:

$$3 \begin{bmatrix} 7 & 5 & -10 \\ 3 & 8 & 0 \end{bmatrix} = \begin{bmatrix} 21 & 15 & -30 \\ 9 & 24 & 0 \end{bmatrix}.$$

Addition / subtraction is done entry-by-entry on corresponding positions:

$$\begin{bmatrix} 5 & 0 & 3 \\ 11 & -1 & 7 \end{bmatrix} + \begin{bmatrix} 1 & -7 & 5 \\ 0 & 3 & -10 \end{bmatrix} = \begin{bmatrix} 6 & -7 & 8 \\ 11 & 2 & -3 \end{bmatrix}.$$

This is **undefined** when the two matrices have different dimensions.

13. Systems of Inequalities and Partial Fractions

13.1 Solving a System of Inequalities

1. Graph each inequality.
2. Plug in the test point $(0,0)$: if it satisfies the inequality, shade the side containing the origin; otherwise shade the other side.
3. Repeat for each inequality — the overlap (intersection) is the system's solution.

13.2 Partial Fraction Decomposition

Used to break a rational expression into simpler fractions (a key setup technique carried into calculus). The general form for distinct linear factors is

$$\frac{1}{(a-b)(b-c)} = \frac{A}{a-b} + \frac{B}{b-c}.$$